CHAPTER 4

Networking, Security & Communications

LEARNING GOALS

- Characteristics of WAN, LAN, MAN, PAN, WPAN
- Client-server & peer-to-peer (P2P) networking models
- Thick client vs. thin client architectures
- Public vs. private cloud computing & storage
- Hardware to support LAN (Hub, Switch, Router, NIC, AP)
- Functions of Routers, Gateways, Bridges, Repeaters
- Data collision detection & avoidance (CSMA/CD, CSMA/CA)
- Differences between Internet & World Wide Web
- Hardware needed to support the Internet & Modems
- Structure of IP Addresses (IPv4 & IPv6)
- Uniform Resource Locator (URL) structure
- Role of the Domain Name Service (DNS)
- Need for communication protocols & protocol stacks
- TCP/IP 4-Layer Model (Application, Transport, Internet, Link)
- Application protocols: HTTP, FTP, POP3, IMAP, SMTP, MIME
- Data security, data privacy, and data integrity
- System protection: Passwords, Firewalls, Antivirus, Digital Signatures
- Security threats: Malware, Hacking, Phishing, Pharming
- Methods to reduce security risks & Access Rights
- Validation checks to ensure data integrity
- Verification methods during data entry & transfer
- Circuit switching mechanics & applications
- Packet switching benefits, drawbacks, & routing
- Symmetric vs. Asymmetric cryptography & Key Distribution
- Plaintext, Ciphertext, Block & Stream Ciphers, Block Chaining
- Digital signatures for verified message origin
- Quantum cryptography & Quantum Key Distribution (QKD)
- SSL and TLS protocols & implementation scenarios
- Acquisition of Digital Certificates & Certificate Authorities

KEY TERMS

ARPAnet: Advanced Research Projects Agency Network; pioneer packet-switching network.

LAN: Local Area Network covering small areas like a single building.

PAN / WPAN: Personal Area Network centred around an individual workspace.

File server: Network server providing centralized data and file access.

Switch: Hardware directing incoming data packets to specific destination addresses.

Modem: Modulator-Demodulator converting analogue and digital signals.

Node: Any device (PC, printer, storage) connected to a network.

Peer-to-peer (P2P): Decentralized network where every node shares resources equally.

Thick client: Device processing data locally independently of a network connection.

Cloud storage: Storing data on remote, off-site servers managed by providers.

Spread spectrum: Wideband radio transmission technique (30m-50m range).

Bridge / Gateway: Bridge links identical protocol LANs; Gateway links different protocols.

Symmetric / Asymmetric: Encryption using single shared key vs. public/private key pairs.

WAN: Wide Area Network covering large geographical regions.

MAN: Metropolitan Area Network covering a city or university campus.

WLAN: Wireless Local Area Network implementing IEEE 802.11 Wi-Fi standard.

Hub: Hardware broadcasting incoming data packets to all connected LAN ports.

Router: Device routing data packets between different subnets/networks.

(W)AP: Wireless Access Point providing wireless node connection to a LAN.

Client-server: Architecture using dedicated servers and client workstations.

Thin client: Device relying on a server for processing power and storage.

Packet: Small discrete block of formatted data sent across a network.

Data redundancy: Storing identical data across multiple servers for fault tolerance.

Frequency hopping: Rapid channel switching across 79 frequencies to prevent noise.

NIC / MAC: Network Interface Card with factory-assigned Hardware Address.

4.1 Network Architecture & Topologies

Computer networks connect computing devices (nodes) to share data, hardware peripherals, and software applications. Networks are primarily classified by their geographical footprint, administrative structure, and processing models.

Geographical Classifications: LAN, WAN, MAN, PAN, WPAN

The scale of a network determines the physical transmission media required, ownership structures, and overall throughput capabilities.

Network Type	Geographic Scale	Ownership & Infrastructure	Data Speeds	Typical Application
PAN / WPAN	Within 10 metres (Personal Space)	Privately owned by individual	High (1–24 Mbps via Bluetooth)	Connecting smartphone to wireless headphones or smartwatch.
LAN (Local Area Network)	Single room, home, or building	Owned entirely by the private organization	Very High (100 Mbps – 10 Gbps)	School computer lab or office Ethernet/Wi-Fi network.
MAN (Metropolitan Area Network)	Across a town, city, or campus	Owned by a consortium or telecom service	High (Fiber backbone, 1–10 Gbps)	Linking multiple university buildings scattered across a city.
WAN (Wide Area Network)	Countries, continents, global	Leased public infrastructure (telecoms, undersea cable)	Variable (Depends on distance/medium)	The global Internet or corporate multi-national network.

> **ANALOGY: UNDERSTANDING GEOGRAPHICAL SCALE**
>
> Think of a **PAN** like a conversation whispered between two people sitting on the same bench. A **LAN** is like an intercom system inside an apartment building—privately wired and extremely fast. A **MAN** is like a city bus system connecting distinct neighborhoods. A **WAN** is the international postal system: it relies on roads, planes, and ships owned by many countries to carry letters around the world.

Client-Server vs. Peer-to-Peer (P2P) Models

The operational logic of a network defines how resources (files, applications, compute power) are managed and accessed by connected nodes.

- **Client-Server Model:** A centralized structure where dedicated, high-performance machines (**servers**) store data, manage security, handle file requests (**File Server**), and manage network access. Client workstations request services, and servers process and respond. Centralized backup, single-point security access, and high scalability are main benefits, though server failure disrupts the whole network.
- **Peer-to-Peer (P2P) Model:** A decentralized structure where every connected **node** acts as both a client and a server. Each node retains its own data and can share files directly with other nodes without a central authority. P2P networks are easy to set up and cost-effective, but security management is fragmented and data redundancy is difficult to control.

File Sharing in P2P: BitTorrent Protocol Terminology

Modern P2P protocols like **BitTorrent** break large binary files into small **pieces** to distribute downloading loads across thousands of computers simultaneously:

- **Tracker:** A central server holding metadata about connected computers without storing actual content files.
- **Swarm:** The entire group of peers sharing or downloading a specific file torrent.
- **Seed:** A peer that has downloaded the complete file and continues uploading pieces to others.
- **Leech:** A node that downloads data from the swarm but limits its upload speed to others.
- **Lurker:** A user who downloads content from a community without contributing any new upload bandwidth.

STUDENT ACTIVITY 4.1: MODEL COMPARISON CHALLENGE

A medium-sized architectural firm with 40 staff members wants to upgrade its network. They work on heavy 3D rendering files and need daily offsite backups and strict user permissions. Should they deploy a **Client-Server** or **Peer-to-Peer** architecture? Justify your recommendation with 3 distinct architectural reasons.

4.2 Client Computations & Cloud Technologies

Thin Client vs. Thick Client Systems

Workstation processing can occur locally on the user's desktop hardware or remotely on centralized enterprise servers.

Attribute	Thin Client	Thick Client
Definition	Low-spec terminal reliant on a network connection to run programs on a central server.	Full-featured PC capable of operating independently offline or online.
Processing Location	Processed completely on the central application server.	Processed locally using internal CPU and RAM hardware.
Storage Location	Remote file servers; no internal local storage required.	Local Solid-State Drive (SSD) or Hard Disk Drive (HDD).
Advantages	Low terminal hardware cost; centralized software updates; easy maintenance.	Works offline; high graphic/compute performance; lower network bandwidth usage.
Disadvantages	Useless without active network/internet connection; high server load.	High initial hardware cost per user; complex software rollout across machines.

ANALOGY: THIN VS. THICK CLIENTS

A **Thin Client** is like a web browser stream of a video game (cloud gaming): your computer sends keypresses and renders video, but the game engine runs on a multi-thousand-dollar server far away. A **Thick Client** is like installing the game directly on your custom gaming rig: it processes everything using your internal graphics card without relying on a constant network connection.

Public and Private Cloud Computing

Cloud storage and cloud computing involve hosting server resources, databases, software applications, and file systems over off-site internet infrastructure. Modern cloud providers maintain extensive **data redundancy** by replicating data across geographic locations to protect against hardware failures.

- **Public Cloud:** Cloud infrastructure owned and operated by third-party service providers (e.g., AWS, Microsoft Azure, Google Cloud). Resources are shared dynamically across millions of multi-tenant users over the public Internet. It provides high scalability and low upfront costs, but offers less control over storage locations.
- **Private Cloud:** Dedicated cloud infrastructure owned, managed, and accessed exclusively by a single enterprise or government agency. It can be hosted on-premise or off-site. It delivers total control, high security, and compliance with data isolation standards, but demands substantial capital expenditure.

4.3 Hardware Required to Support a LAN

Building a physical or wireless LAN requires specialized communication hardware to convert, connect, route, and amplify network signals.

- **Network Interface Card (NIC):** Hardware controller installed in a device that allows it to connect to a network. Each NIC contains a factory-embedded physical address called a **MAC (Media Access Control) address**.
- **Hub:** A basic hardware device that connects multiple network nodes. It broadcasts incoming data frames to *every single port* on the hub, creating unnecessary network traffic and security risks.
- **Switch:** An intelligent LAN device that reads the destination MAC address of incoming data frames and forwards them *only to the specific target port*.
- **Wireless Access Point ((W)AP):** A device containing a transceiver that allows wireless nodes (smartphones, laptops) to connect to a wired LAN using radio frequencies.
- **Repeating Hub / Repeater:** A device that receives weak incoming signals, cleans noise, amplifies the voltage, and retransmits them to overcome cable distance limits.
- **Bridge:** A networking device used to join two separate LAN segments that run on the *same communications protocol*.
- **Gateway:** A specialized entry point that translates data format protocols between two networks running on *different communication protocols*.
- **Modem (Modulator-Demodulator):** Converts digital computer data into analogue signals for transmission over telephone lines, and converts incoming analogue signals back to digital data. Softmodems perform these signal conversion steps in software to save hardware component costs.

WORKED EXAMPLE 4.1: Switch vs. Hub Bandwidth Efficiency

Scenario: Node A sends a 10 MB file to Node B on an 8-port hub vs. an 8-port switch.

Hub Behavior: Broadcasts all 10 MB of data frames across ports 2 through 8. Nodes C, D, E, F, G, and H receive the frames, inspect the destination MAC address, and drop them. Bandwidth is consumed across the whole network segment.

Switch Behavior: Reads destination MAC address in the frame header, checks its internal MAC table, opens a direct circuit between Port A and Port B, and transmits data solely to Node B. Other ports remain free for concurrent communications.

4.4 Transmission Media & Collision Handling

Physical and Wireless Transmission Media

- **Twisted Pair Cable:** Copper wires twisted together in pairs to reduce electromagnetic interference (crosstalk). Common standard in Ethernet networks (Cat5e/Cat6).
- **Wi-Fi (IEEE 802.11):** Uses high-frequency radio and microwave bands for wireless LAN communication.
- **Bluetooth:** Short-range wireless connectivity using radio waves in the 2.45 GHz ISM band, typically for WPAN setups.
- **Spread Spectrum Frequency Hopping:** A radio transmission method where a device rapidly hops across 79 random frequency channels. If a chosen channel has interference, it immediately switches to another channel, allowing robust short-range transmissions up to 100 metres.

Data Collisions on Shared Media

When multiple nodes transmit data simultaneously over a shared medium (such as a single copper wire or shared radio spectrum), signals overlap and corrupt each other. This is known as a **data collision**.

> **ANALOGY: CARRIER SENSE & COLLISION HANDLING**
>
> Imagine a dinner party in a dark room. **CSMA/CD** is like two people starting to talk at the exact same moment. They both stop, wait a random number of seconds, and then try speaking again. **CSMA/CA** is like raising your hand and waiting for the host's clear signal before speaking to avoid interrupting anyone.

CSMA/CD vs. CSMA/CA

- **CSMA/CD (Carrier Sense Multiple Access with Collision Detection):** Used primarily in wired Ethernet networks.
  1. Node listens to the medium to check if line is idle.
  2. If idle, node begins transmitting data packet.
  3. Node monitors the voltage level on the wire during transmission.
  4. If a sudden rise in voltage indicates a collision, a **jamming signal** is sent to alert all nodes.
  5. Transmission halts immediately, and both nodes wait a random time interval (backoff timer) before retrying.
- **CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance):** Used in wireless Wi-Fi networks where nodes cannot listen while transmitting.
  1. Node checks channel; if clear, it sends a **Request to Send (RTS)** packet to the access point.
  2. Access Point replies with a **Clear to Send (CTS)** packet, reserving the channel.
  3. Node transmits its payload and waits for an **ACK (Acknowledgement)** frame. If ACK is missing, a collision is assumed and backoff occurs.

4.5 Internet Infrastructure, URLs & DNS

The **Internet** is the global physical infrastructure of interconnected computer networks. The **World Wide Web (WWW)** is an information system of interlinked web pages and resources accessed *over* the Internet using HTTP protocol.

Structure of a Uniform Resource Locator (URL)

A URL provides the precise location address of a Web resource across the Internet:

URL BREAKDOWN STRUCTURE

`https://www.example.com:443/courses/computer-science/notes.html#section2`

- **Protocol (Scheme):** `https://` (Hypertext Transfer Protocol Secure)
- **Domain Name (Host):** `www.example.com` (Human-readable server name)
- **Port Number:** `:443` (Target application port; default for HTTPS)
- **Path:** `/courses/computer-science/` (Directory path on web server)
- **Resource / Filename:** `notes.html` (Target document file)
- **Fragment / Anchor:** `#section2` (Specific position marker within the file)

The Domain Name System (DNS) Translation Process

Computers route data using numeric **IP addresses** (e.g., `192.0.2.1`), but humans remember domain names (e.g., `cambridge.org`). The DNS converts domain names into destination IP addresses through a recursive step-by-step lookup process:

1. User enters domain name into a browser. Local OS cache is checked first.
2. If missing, request is sent to the local ISP's **DNS Recursive Resolver**.
3. Resolver queries a **Root Name Server**, which directs it to the appropriate **Top-Level Domain (TLD) Server** (e.g., .org, .com).
4. TLD server returns the IP address of the domain's **Authoritative Name Server**.
5. Authoritative server returns the target IP address to the resolver.
6. Resolver caches the result and returns IP address to client browser to establish TCP connection.

4.6 Protocols & The TCP/IP Protocol Stack

A **protocol** is a set of formal rules governing communication across a network. Without agreed rules for framing, addressing, error detection, and speed, nodes cannot interpret received bitstreams.

The 4-Layer TCP/IP Protocol Stack

The TCP/IP stack structures network operations into four modular layers. As data moves down the stack from sender to recipient, each layer adds specific header metadata (**encapsulation**).

Layer	Protocol Data Unit (PDU)	Core Responsibilities	Key Protocols
1. Application Layer	Data / Message	Provides user-facing network services and high-level application data formatting.	HTTP, HTTPS, FTP, SMTP, POP3, IMAP, DNS
2. Transport Layer	Segment (TCP) / Datagram (UDP)	End-to-end host-to-host communications, flow control, port numbers, packet reassembly, error recovery.	TCP, UDP
3. Internet Layer	IP Packet	Logical addressing (IP addresses) and routing data packets across disparate networks.	IP (IPv4, IPv6), ICMP, ARP
4. Link Layer	Ethernet Frame	Physical hardware drivers, MAC addressing, signal encoding, and physical transmission.	Ethernet, Wi-Fi (802.11), PPP

ANALOGY: TCP/IP ENCAPSULATION AS POSTAL MAILING

1. **Application:** You write a letter (Data payload).
2. **Transport:** You break a long letter into numbered pages and put them in envelopes with sequence numbers (TCP Segment).
3. **Internet:** You write destination and return street addresses on the envelope (IP Packet).
4. **Link:** Postal workers place envelopes inside specific mail trucks moving over physical roads (Ethernet Frame).

Application Protocols Deep-Dive

- **HTTP / HTTPS:** Transfers hypertext documents across the web. HTTPS encrypts data using SSL/TLS.
- **FTP (File Transfer Protocol):** Used for uploading and downloading binary and text files between client and server.
- **Email Protocols (Push vs. Pull):**
  - **SMTP (Simple Mail Transfer Protocol):** A **push protocol** used by clients to upload emails to outgoing mail servers and for servers to relay messages between each other. Handles **MIME (Multipurpose Internet Mail Extensions)** data to send binary attachments (images, audio).
  - **POP3 (Post Office Protocol v3):** A **pull protocol** that downloads emails from server to local client and deletes server copies by default.
  - **IMAP (Internet Message Access Protocol):** A **pull protocol** that syncs email client headers with remote server folders, keeping emails cached on server across multiple client devices.

STUDENT ACTIVITY 4.2: PROTOCOL IDENTIFICATION

Identify which protocol is actively used in each scenario:
a) Fetching unread emails on a laptop and keeping them visible on a smartphone inbox simultaneously.
b) Sending a PDF file attachment via email over an encrypted channel.
c) Uploading a 2 GB file to a remote website storage folder via command line terminal.

4.7 Network Security, Privacy & Data Integrity

Managing network safety requires separating three distinct domain objectives:

- **Data Security:** Protection of computer systems and data from unauthorized access, accidental loss, malicious deletion, or physical damage.
- **Data Privacy:** Rights of individuals/organizations to control how their personal data is collected, stored, shared, and processed by third parties.
- **Data Integrity:** Ensuring that data remains accurate, complete, authentic, and uncorrupted during entry, storage, and transmission over network channels.

Common Security Threats

Threat Category	Mechanism & Operational Method	Mitigation Strategies
Viruses & Malware	Self-replicating malicious code that attaches to host files and executes unauthorized operations.	Antivirus software, regular OS updates, restricting executable downloads.
Spyware & Keyloggers	Hidden software tracking keystrokes, activity, and credentials, relaying them secretly to an **eavesdropper**.	Anti-spyware utilities, multi-factor authentication (MFA), virtual onscreen keyboards.
Hacking	Unauthorized access to digital systems by exploiting software vulnerabilities or weak access points.	Network firewalls, strong password policies, intrusion detection systems (IDS).
Phishing	Social engineering attack using fake emails/websites designed to trick users into revealing login details.	User security training, email spam filters, digital signatures on domain mail.
Pharming	Malicious tampering with DNS hosts or browser settings to redirect requests to fraudulent mirror sites automatically.	DNSSEC verification, updating host files, inspecting SSL certificates.

System Protection Mechanisms

- **User Accounts & Passwords:** Access control mechanisms enforcing authentic identity verification and user access rights.
- **Firewalls:** Hardware or software filters examining incoming and outgoing traffic packets against predefined rule sets, blocking unauthorized network ports and IP addresses.
- **Validation:** Automated software checks verifying input data matches mandatory criteria (e.g., range checks, format checks, length checks) *before* saving to preserve data integrity.
- **Verification:** Methods ensuring data entered or transferred matches the original source exactly without errors (e.g., double entry checks, parity bits, checksums).

WORKED EXAMPLE 4.2: Validation vs. Verification

Validation: Inputting a birthdate field `31/02/2024` fails a *Format Check / Logic Check* (February does not have 31 days). Validation confirms data is reasonable and formatted correctly.

Verification: Inputting password `Secr3t!` twice into registration boxes confirms double-entry match. Verification checks that data was transcribed accurately, even if the value itself was typed incorrectly in both steps.

4.8 Switching Mechanisms: Circuit vs. Packet Switching

Circuit Switching Mechanics

Circuit switching establishes a dedicated, unbroken physical communication path between sender and receiver for the entire duration of a call or session.

- **Steps:** 1. Circuit Setup → 2. Data Transmission → 3. Circuit Teardown.
- **Advantages:** Dedicated bandwidth ensures low latency, zero jitter, and constant transmission speeds—ideal for real-time video conferencing or landline phone calls.
- **Drawbacks:** Inefficient channel usage; bandwidth is reserved even during silent pauses, blocking other traffic.

Packet Switching Mechanics

Packet switching splits messages into small discrete blocks called **packets** that travel independently across network nodes based on dynamic routing decisions.

ANATOMY OF A NETWORK PACKET

Header Metadata	Payload	Trailer / Footer
• Source IP & MAC Address • Destination IP & MAC Address • Packet Sequence Number • Protocol Identifier	Actual file data fragment / message slice (e.g., 1500 bytes of image data).	• Error checking checksum (e.g., Cyclic Redundancy Check - CRC) • End of packet flag

Routing & Packet Reassembly

Network **routers** inspect destination addresses in packet headers and forward packets dynamically along the shortest or least congested path available. Packets may arrive out of order; the receiving node's Transport layer uses sequence numbers to reorder and reassemble the original file.

Metric	Circuit Switching	Packet Switching
Path Reservation	Dedicated fixed physical path for entire session.	Dynamic routing per packet; paths vary dynamically.
Bandwidth Efficiency	Low; reserved capacity wasted during idle moments.	High; multiple nodes share backbone links concurrently.
Latency & Reliability	Low fixed latency; session fails if connection breaks.	Variable latency (jitter); robust against node outages.

4.9 Cryptography, Quantum Security & Digital Certificates

Symmetric vs. Asymmetric Encryption

Encryption transforms original readable message text (**plaintext**) into an unreadable mathematical format (**ciphertext**) using an encryption algorithm and key.

- **Symmetric Encryption:** Uses the *same single secret key* for both encryption and decryption. Fast and computationally efficient, but suffers from the **Key Distribution Problem**—sending the shared key securely over public networks to the recipient risks interception by an **eavesdropper**.
- **Asymmetric Encryption:** Uses a mathematically linked key pair: a **Public Key** (known to everyone) and a **Private Key** (kept secret by owner). Messages encrypted with a recipient's Public Key can *only* be decrypted by that recipient's Private Key.

WORKED EXAMPLE 4.3: Digital Signature Generation & Verification

Goal: Alice sends a document to Bob, proving it originated from her and was not modified.

1. Alice generates a cryptographic hash of the plaintext message.
2. Alice encrypts the hash using her own **Private Key**. This encrypted hash is the **Digital Signature**.
3. Alice transmits the message and digital signature to Bob.
4. Bob decrypts the signature using Alice's **Public Key** to reveal the original hash.
5. Bob hashes the received message independently. If Bob's hash matches the decrypted hash, message origin and integrity are confirmed.

Stream Ciphers vs. Block Ciphers & Block Chaining

- **Stream Cipher:** Encrypts data bit-by-bit or byte-by-byte continuously in sequence as data arrives.
- **Block Cipher:** Processes data in fixed-size blocks (e.g., 64-bit or 128-bit blocks) in one go.
- **Block Chaining (CBC):** Advanced cipher mode where each plaintext block is combined with the previous block's ciphertext (using an XOR operation) before encryption. This prevents identical plaintext blocks from generating identical ciphertext blocks.

SSL/TLS and Digital Certificates

Secure Sockets Layer (SSL) and its successor **Transport Layer Security (TLS)** protect web sessions (e.g., online banking, e-commerce). A website acquires a **Digital Certificate** from a trusted **Certificate Authority (CA)** to prove its domain ownership and supply its public key.

Quantum Cryptography & Quantum Key Distribution (QKD)

Traditional asymmetric encryption relies on mathematical problems (like prime factorisation) that could be broken by future quantum computers. **Quantum Key Distribution (QKD)** uses quantum photon states to transmit encryption keys. According to quantum physics (Heisenberg Uncertainty Principle), any eavesdropper attempting to measure a photon alters its state instantly, revealing the intrusion and invalidating the key.

STUDENT ACTIVITY 4.3: SECURITY SCENARIO ANALYSIS

Evaluate which security mechanism best solves each problem:
1. Preventing hackers from decrypting captured Wi-Fi traffic.
2. Guaranteeing an email contract received from a vendor was actually sent by them and untouched in transit.
3. Stopping employees from opening malicious websites on internal office computers.

4.10 End-of-Chapter Assessment Exercises

EXERCISE QUESTIONS

1. Network Fundamentals & Architectures
a) Contrast a Local Area Network (LAN) and a Wide Area Network (WAN) in terms of geographic scale, ownership, and transmission speeds. [3 marks]
b) Compare the client-server model with the peer-to-peer (P2P) model regarding resource management and scalability. [3 marks]
c) Differentiate between a thick client and a thin client based on where data processing and storage take place. [2 marks]
d) What are the primary trade-offs (in terms of cost, security, and control) between using a public cloud versus a private cloud? [3 marks]

2. Medium Access & Collisions
How do network systems handle data collisions when multiple devices attempt to transmit data simultaneously over a shared medium (e.g., using CSMA/CD or collision avoidance techniques)? [4 marks]

3. Web Infrastructure & Addressing
a) Break down the distinct parts of a Uniform Resource Locator (URL) (e.g., protocol, domain, path) and explain how it locates a specific resource. [3 marks]
b) Describe the step-by-step role of the Domain Name Service (DNS) when translating a human-readable URL into an IP address. [4 marks]

4. Protocol Stacks & Applications
a) Outline the four layers of the TCP/IP protocol stack in correct order (from top to bottom or bottom to top), giving the core responsibility of each layer. [4 marks]
b) Briefly state the specific purpose of each of the following application/transport protocols: HTTP, FTP, POP3, and IMAP. [4 marks]

5. Switching Paradigms
a) Explain what is meant by circuit switching. [2 marks]
b) There are many applications in which digital data are transferred across a network. Video conferencing is one of these. For this application, circuit switching is preferred. Explain why this is so. [2 marks]
c) List two major benefits and one drawback of packet switching. [3 marks]

6. Data Security, Validation & Verification
a) Differentiate between the concepts of data security, data privacy, and data integrity. [3 marks]
b) Contrast the use of validation (e.g., range checks, format checks) with verification (e.g., double-entry, parity checks) in protecting data during entry and transfer. [4 marks]
c) Name three mechanisms used to protect computer systems against threats. [3 marks]

7. Encryption Mechanics & Certificates
a) Compare symmetric key cryptography with asymmetric key cryptography in terms of key management, speed, and how plaintext is converted to ciphertext. [4 marks]
b) Explain how an asymmetric key pair can be used in reverse to generate a digital signature that verifies the sender's identity and ensures message integrity. [3 marks]
c) What are SSL and TLS, and in what real-world scenarios are they typically deployed to secure network traffic? [3 marks]

CHAPTER 4 SELF-ASSESSMENT CHECKLIST

- [ ] I can contrast LAN, WAN, MAN, PAN, and WLAN networks.
- [ ] I can explain Client-Server, P2P, Thin Client, and Thick Client models.
- [ ] I can outline hardware roles: Routers, Switches, Hubs, NICs, Gateways, Bridges, and Modems.
- [ ] I know how CSMA/CD and CSMA/CA handle network data collisions.
- [ ] I can dissect a URL and trace a step-by-step DNS lookup.
- [ ] I understand the 4 layers of TCP/IP and email push/pull protocols (SMTP, POP3, IMAP).
- [ ] I can contrast Circuit Switching with Packet Switching.
- [ ] I can explain Symmetric vs. Asymmetric encryption, Digital Signatures, QKD, and SSL/TLS.